KLDet: Detecting Tiny Objects in Remote Sensing Images via Kullback–Leibler Divergence

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***Abstract*— Remote sensing images (RSIs) frequently contain quite a few tiny objects with a finite number of pixels to study. The limited spatial information poses a challenge for extracting discriminative features for representing the characteristics of tiny objects. Existing solutions mainly focus on aggregating contextual information at different levels, while rarely touching the step that is crucial for model training, i.e., label assignment. Tiny instances occupy fairly small regions of images and have limited overlaps to priors (anchors or dots), which is a dilemma for traditional label assignment strategies. Despite being simple and effective, the mainstream Intersection over Union (IoU)-based label assignment strategy struggles to accurately measure the localization of tiny bounding boxes. In contrast, the Kullback–Leibler divergence (KLD) localization metric accurately reflects minor offsets of tiny bounding boxes. More importantly, KLD is able to measure non-overlapping bounding boxes, providing an advantage in mining more potential positive samples of tiny objects. In this article, from a cost-efficient point of view, we detect tiny objects through KLD in the form of single-stage framework. Specifically, we model the parameterized bounding box as a 2-D Gaussian distribution (Bbox2Gaussian) in order to use KLD as a localization metric. Then, we propose an adaptive online training sample mining (Ali-TSM) strategy based on inter-distribution similarity, which selects high-quality positive samples by considering localization and classification rather than just centroid distance or IoU. Finally, task-level attention (TlA) is introduced to guide the model in freely selecting the appropriate features for the classification or regression task. We conducted extensive experiments on four popular public datasets. Compared to the baseline, KLDet improves performance on Tiny Object Detection in Aerial Images (AI-TOD) and object DetectIon in Optical Remote sensing image (DIOR) by 4.1 AP and 6.7 mAP. On VisDrone and Small Object Detection dAtasets (SODA-D), KLDet exhibits superior performance than baseline. The code is available at *https://github.com/TinyOD/mmdet-kldet*.**

***Index Terms*— Kullback–Leibler divergence (KLD), label assignment, object detection, remote sensing images (RSIs), tiny objects.**

I. Introduction

THE task of remote sensing object detection is to predict the pre-defined categories and locations of objects in the region of interest. Compared to natural images, remote sensing images (RSIs) are usually captured in a larger field of view, which makes them susceptible to interference from complex backgrounds and suffer from blurred appearance, occlusion, and dense alignment [1]. Meanwhile, tiny objects often appear in RSIs, and they are usually represented in a limited number of pixels (less than 16×16 in Tiny Object Detection in Aerial Images (AI-TOD) [2]). They are challenging to extract features and cannot be adapted to mainstream label assignment strategies, which seriously hinders further advancement of remote sensing detection accuracy [3].

Recently, numerous methods have been proposed to address the above problems. MSFC-Net [4] introduces compound semantic feature fusion in a single-stage detector to generate semantic descriptions for small objects in complex backgrounds. Yu and Ji [5] introduce deformable convolutional network (DCN) [6] to accommodate the scale variations of different object instances. Qin et al. [7] propose a feature fusion strategy to exploit the feature representations, and introduce dilated convolution to obtain context to support small object detection.

Despite these methods being well-validated in practice, they usually involve complex feature fusion strategies or costly sampling paradigms (e.g., DCN). Besides, they do not touch the key point of training models, i.e., label assignment. In other words, the anchor assigned to each ground truth (GT) may be suboptimal as Intersection over Union (IoU) cannot accurately measure the offset of tiny objects [3]. More importantly, the mainstream label assignment strategy based on predefined metrics leads to imbalanced assignment results, which makes the model focus more on normal objects while ignoring tiny objects. Finally, the *Head* of the above method still takes the output of feature pyramid network (FPN) [8] as its input directly, which leads the model to lack support for different tasks (i.e., classification and regression) [9], especially for tiny objects that need to be located precisely.

In light of above findings, we reconsider the label assignment in remote sensing detection and build a novel single-stage detector called KLDet. Specifically, KLDet contains the following three components.

A. Bounding Box as a 2-D Gaussian Distribution

Inspired by nature observation, the foreground and background pixels of tiny objects are concentrated in the center and boundary of bounding boxes [10]. To better represent the distribution of tiny objects on feature maps, we transform the

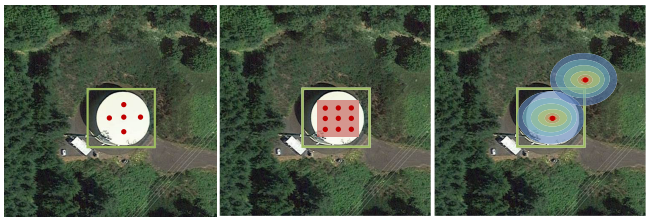
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(a) (b) (c)

Fig. 1. Different sampling strategies in dense object detection. The green box is GT. The solid red circle represents the positive sample. (a) IoU-based sampling strategies primarily rely on predefined thresholds and require high accuracy in anchor localization. (b) Region-based sampling strategy selects positive and negative labels by judging whether the anchors fall within a fixed range of GT. (c) Our KLDet samples the training set based on the instance-wise distribution, which is more friendly to small objects. Best viewed in color.

parameterized bounding box into a 2-D Gaussian distribution with decreasing probability from the centroid to the boundary. Unlike IoU or region-based localization metrics that cannot measure non-overlapping bounding boxes [Fig. 1(a) and (b)], we propose to use Kullback–Leibler divergence (KLD) [11] to evaluate localization quality [Fig. 1(c)]. In this way, the label assignment results will not be influenced by the instance's area, thus avoiding the limitations of traditional assignment strategies that tend to favor larger-sized objects. Furthermore, KLD can also measure non-overlapping bounding boxes, which is quite common during the training process of tiny object detectors [12].

B. Adaptive Online Training Sample Mining

In order to select high-quality training samples, we import online information (i.e., predicted classification confidence and coordinates) in the label assignment. Existing optical remote sensing object detectors typically use offline information (e.g., pre-defined IoU or fixed areas of GT center) to determine the boundary between positive and negative samples. While simple and effective, they ignore the fact that the optimal boundaries for objects with different shapes, sizes, and occlusion conditions may differ. We suggest using online information as dynamic bounds to adaptively identify high-quality positive samples. Such a rigid split conditioned on pre-defined metrics is broken by the online information, allowing difficult-to-locate tiny objects to easily obtain sufficient training samples.

C. Task-Level Attention

To support tiny objects that require precise localization, we introduce a switchable function called task-level attention (TlA). Due to the success of the RetinaNet [13] architecture, the parallel *Head* has become predominant in the recent past. However, studies [9], [14] have already shown that *Head* branches corresponding to different tasks prefer different feature patterns, while current remote sensing detection ignores this fact. To alleviate this issue, we propose TlA, which ensures that the two tasks adaptively seek the best position in the channel without compromising with each other.

The proposed components can collaboratively work to output high-quality classification and localization results. Extensive experiments are carried out on four challenging tiny object datasets, such as AI-TOD [2], object DetectIon in Optical Remote sensing image (DIOR) [15], VisDrone [16], and Small Object Detection dAtasets (SODA-D) [17]. With negligible extra parameters, our KLDet outperforms the latest single-stage detectors by a clear margin of 3.3 AP on AI-TOD. Results on DIOR show that KLDet achieves state-of-the-art performance with 73.5 mAP. Besides, KLDet outperforms existing methods on VisDrone and SODA-D with a clear margin, especially for detection accuracy on tiny objects.

The main contributions of this work can be summarized as follows.

1) Our work introduces KLDet, a simple, fast, and accurate framework for detecting tiny objects in RSIs. It focuses on improving detection performance through well-designed label assignment strategies, providing a cost-efficient alternative to feature enhancement-based methods.
2) To address the challenge of localizing tiny objects, we transform parameterized bounding boxes into 2-D Gaussian distributions. Through using the similarity between distributions (KLD) as a localization metric, our method outperforms IoU in measuring minor offsets of tiny boxes and mining more positive samples.
3) We propose an adaptive online training sample mining (Ali-TSM) strategy to dynamically determine positive and negative sample boundaries based on online information. Such an approach mitigates imbalanced assignment results caused by rigid division conditional on IoU.
4) We introduce TlA with a switch function, which enables a flexible selection of channel features to better support classification and regression tasks.

The rest of this article is organized as follows. Section II briefly reviews the related work. Section III introduces the three components of KLDet. Comprehensive experimental and comparative details are reported in Section IV. Finally, Section V provides the conclusion of this article.

II. Related Work

A. Object Detector for RSIs

With the remarkable success of generic detectors in large-scale image processing, it has also swept the remote sensing community. Detectors employed for RSIs can be categorized into two groups including two-stage and one-stage.

The two-stage framework defines the object detection task as a step-by-step refining, i.e., regional proposals and bounding boxes refinements [18]. ABNet [1] introduces channel attention to improve the feature representation of the backbone to mitigate the obstruction of foreground objects by complex backgrounds. SCRDet++ [19] performs instance-level denoising of feature maps to enhance the detection of small objects. CBD-E-FPN [20] proposes a contextual bidirectional enhancement method to remove unexpected background. CSFF [21] models the relationship between different feature channels to obtain a powerful and discriminative multilevel feature representation. While these methods enjoy high accuracy, the

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complex refinement operations make them bear high computational costs.

The single-stage framework performs detection as a single step, i.e., predicts both classification and location. HawkNet [22] utilizes an up-scale feature aggregation framework and a novel up-sampling method to differentiate small object features. FSoD-Net [23] contains a multiscale augmentation network and three different isolated regression branching layers to correspond to small-, medium-, and large-scale objects. ASSD [24] proposes a context-based feature alignment module to correct misalignments between kernels and objects and extract aligned and robust features. O^2-DNet [25] represents oriented objects as paired middle lines and eliminates pre-designed anchor boxes to better match the aspect ratio of small bounding boxes. Yang et al. [26] model the rotated bounding box as a 2-D Gaussian distribution (Bbox2Gaussian), and then calculate the KLD between the Gaussian distributions as a regression loss to adjust the rotation angle. Despite attracting a surge of research interest, the aforementioned methods usually require constructing complex semantic descriptions for pixel-level centroids. This is inconsistent with the initial design concept of simple and fast single-stage object detectors. Yang et al. [26] mention KLD but does not explore its application in label assignment.

Recently, Transformer [27] has rapidly gained popularity in computer vision, especially for object detection. Deformable-DETR [28] significantly enhances the detection of small objects via effectively preserving the spatial resolution by multiscale deformable attention modules. DINO [29] enhances the stability of the decoder by improving the label assignment and impressively boosts performance for small objects. DAB-DETR [30] further improves detection accuracy with dynamically adjustable anchor boxes. However, such prosperity cannot camouflage the unsatisfactory detection of tiny objects by transformer-based detectors, as they struggle to model local information.

In this work, we re-examine the localization metrics suitable for tiny objects, and allocate sufficient and high-quality positive samples for them using an elegant and efficient strategy.

B. Label Assignment in Object Detection

The task of selecting which anchors are to be assigned as positive or negative samples is considered to be a crucial factor that greatly affects the detectors' performance [13], [31], [32]. Label assignment can be divided into hard and soft strategies according to whether the split conditions of training samples are fixed or not.

For the hard label assignments, the classical anchor-based detectors [13], [18], [33], [34] assign positive samples by calculating the IoU between proposal and GT. The anchor-free detectors [31], [35], [36] treat the anchors that appear in a fixed region in the center of GT as positive samples. However, the use of pre-defined thresholds is simply based on ad hoc heuristics, and more importantly, the resulting hard division of anchors as either positive or negative samples is questionable [37].

Recently, researchers attempted to construct a flexible splitting boundary to train detectors efficiently. ATSS [38] utilizes a linear combination of IoU's statistical values as the division threshold. NoisyAnchor [37] and VFNet [39] assign soft labels to the training samples to mitigate the noise generated by the binary labels. These excellent works inspire us, but their performance on tiny object detection tasks commonly suffers a precipitous decline.

In RSIs detection, some works try to develop label assignment strategies that are friendly to tiny objects. NWD [3] uses a fixed threshold to assign training samples based on Gaussian Wasserstein distance (GWD) [40]. After that, RFLA [41] sorts the anchors according to the similarity between distributions (i.e., anchors and GTs), and takes the top k as positive samples. However, these schemes either retain rigid division conditions, or only consider the localization quality of anchors and ignore the effect of classification accuracy. In this work, we develop label assignment strategies suitable for tiny objects in terms of accurate localization metrics and flexible splitting conditions.

C. Attention Mechanism in FPN

The success of attention mechanisms in machine translation has motivated researchers to explore them extensively in the computer vision community [42]. SENet [43] adopts global average pooling (GAP) to estimate the channel weights, and then uses multilayer perceptron (MLP) to complete the nonlinear mapping of weights to obtain the relationships of different channels in ResNet [44]. After that, CBAM [45] introduces spatial attention based on SENet. ECANet [46] outperforms CBAM with fewer parameters by 1-D convolution instead of fully connected layers.

Inspired by the above mechanism, researchers transferred attention mechanisms to FPN to support RSIs. ABNet [1] yields balanced multiscale features by channel and spatial attention to better extract features from dense objects in FPN. CADNet [47] introduces spatial-and-scale-aware attention modules to optimize the region proposal network. Yu and Ji [5] enhance multiscale representation by an attention module embedded in a multilayer neuron feature extractor.

The methods mentioned above mainly focus on obtaining discriminative features while ignoring the coordination with the detector's *Head*. This approach limits further growth in gain, as *Head* prefers different feature patterns for classification and regression tasks [9]. In contrast to them, we propose TlA with a switch function to ensure that the detector freely selects the appropriate feature pattern, thus supporting the need for precise localization of tiny objects.

III. Proposed Methodology

The framework of KLDet is shown in Fig. 2. It models the parameterized Bbox2Gaussian. Then, the training samples are sampled by Ali-TSM. Besides, TlA is used to select appropriate features for classification and regression tasks. The details of the three components and loss function are described below.

A. Bounding Box as a 2-D Gaussian Distribution

Modeling the tiny Bbox2Gaussian is inspired by nature insight, where the foreground and background of tiny objects

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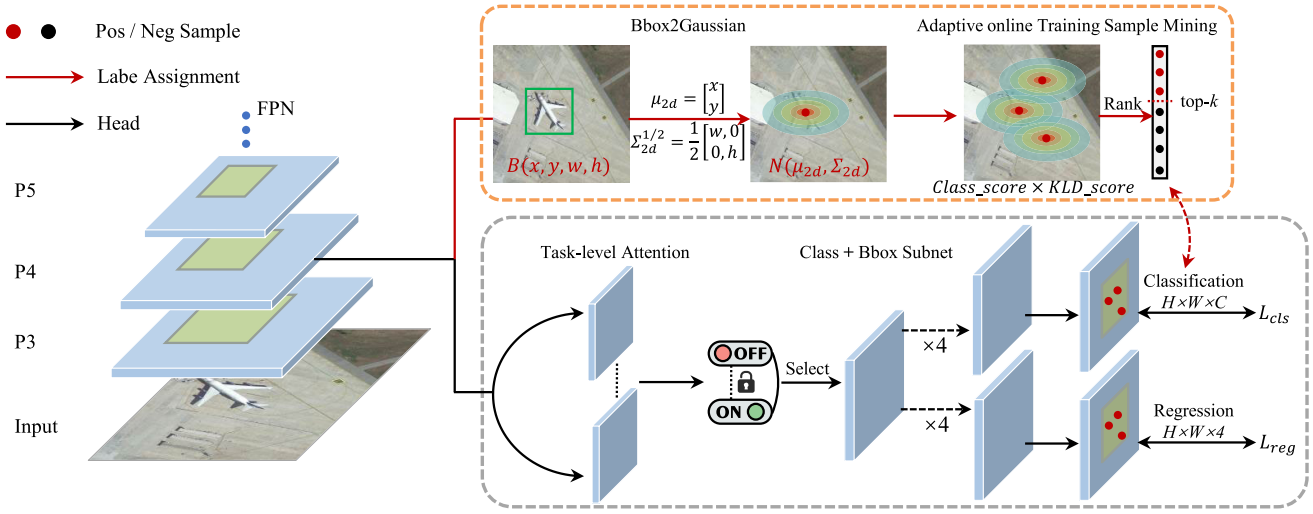


Fig. 2. Architecture of our KLDet. In Bbox2Gaussian, the distribution parameters (μ_{2d}, Σ_{2d}) denote the location and scale of the box, where $_{2d}$ denotes two dimensional. In Ali-TSM, the red double-arrowed dotted line indicates that the ranking is based on the product of classification confidence (Class_score) and localization metric (KLD_score). In TIA, "lock" means selecting appropriate channel features according to certain rules, thus assisting the model to accurately classify or regress. Details of the rule are shown in Fig. 5. Note that our sampling strategy is completely cost-free on the extra parameters. Best viewed in color.

are concentrated at the center and boundary of the box, respectively [10]. That is, the importance of the pixels inside the box decreases in order from the center to the border. To clearly describe this property, a 2-D Gaussian distribution is an intuitive solution. In this part, we first prove that the projection of the 2-D Gaussian distribution on the 2-D plane is an ellipse. Subsequently, it is considered as a maximal interior ellipse of the bounding box, i.e., the 2-D Gaussian distribution is related to the bounding box.

The probability density function of a 2-D Gaussian distribution is as follows:

$$f(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma})=\frac{1}{2\pi|\Sigma|^{\frac{1}{2}}}\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right) \quad (1)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ denote the mean vector and covariance matrix of 2-D Gaussian distribution, respectively. When $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are fixed, (1) is only related to $\boldsymbol{x}$

$$\Delta=(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) \quad (2)$$

since $\boldsymbol{\Sigma}$ is a positive definite matrix, the following eigenvalue decomposition can be performed:

$$\begin{aligned}&\boldsymbol{\Sigma}=\boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^T,\quad \boldsymbol{U}\boldsymbol{U}^T=\boldsymbol{I}\\&\boldsymbol{\Lambda}=\mathrm{diag}(\lambda_i),\quad i=1,2,\ldots,d,\ \boldsymbol{U}=[\boldsymbol{u}_1,\boldsymbol{u}_2,\ldots,\boldsymbol{u}_d]_{d\times d}\end{aligned} \quad (3)$$

where d denotes the dimension of Gaussian distribution and $\boldsymbol{\Sigma}$ can be reformulated as

$$\boldsymbol{\Sigma}=\sum_{i=1}^{d}\boldsymbol{u}_i\lambda_i\boldsymbol{u}_i^T. \quad (4)$$

Equation (2) can be reformulated as

$$\Delta=\sum_{i=1}^{d}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{u}_i\frac{1}{\lambda_i}\boldsymbol{u}_i^T(\boldsymbol{x}-\boldsymbol{\mu}) \quad (5)$$

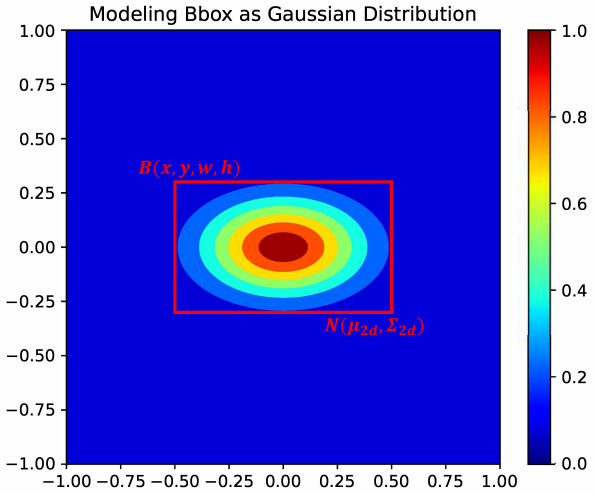


Fig. 3. Illustration of modeling a horizontal Bbox2Gaussian.

let $y_i=(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{u}_i$. Then, (5) can be reformulated as

$$\Delta=\sum_{i=1}^{d}\frac{y_i^2}{\lambda_i} \quad (6)$$

consider the special case of $d=2$

$$\Delta=\frac{y_1^2}{\lambda_1}+\frac{y_2^2}{\lambda_2}. \quad (7)$$

It can be seen that the projection of 2-D Gaussian distribution in Y_1OY_2 plane is an ellipse when Δ is fixed, whose long and short axes are related to λ_i. From (7), the mean vector and the eigenvalues of the covariance control the position and scale of the ellipse, respectively (as shown in Fig. 3). Therefore, the horizontal bounding box $\mathrm{B}=\{x,y,w,h\}$ (central point coordinates, width and height) [13], [18] can be modeled as a

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2-D Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \boldsymbol{\Sigma}^{1/2} = \frac{1}{2}\begin{bmatrix} w & 0 \\ 0 & h \end{bmatrix}. \tag{8}$$

Although previous studies [3], [40] used GWD [48] to measure the distance between distributions, the lack of scale invariance limits its generalization. In this work, we use a classical statistical distance, KLD, to measure the difference between probability distributions. The closed-form solution of the KLD between two Gaussian distributions $\mathcal{N}_p(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$ and $\mathcal{N}_t(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ is as follows [11]:

$$D_{kl}\left(\mathcal{N}_p||\mathcal{N}_t\right) = \frac{1}{2}\left(\boldsymbol{\mu}_p - \boldsymbol{\mu}_t\right)^T \boldsymbol{\Sigma}_t^{-1}\left(\boldsymbol{\mu}_p - \boldsymbol{\mu}_t\right) + \frac{1}{2}\boldsymbol{Tr}\left(\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Sigma}_p\right) + \frac{1}{2}\ln\frac{|\boldsymbol{\Sigma}_t|}{|\boldsymbol{\Sigma}_p|} - 1 \tag{9}$$

where p and t denote the predicted box and GT, respectively. Through calculating the KLD, we can quantify the degree of difference between two 2-D Gaussian distributions. A smaller KLD indicates that the two distributions are similar and the boxes can be considered to be closer together. Notably, [26] has demonstrated that the KLD has scale invariance between 2-D Gaussian distributions. It implies that KLD has a greater potential for application because cross-scale objects are widely available in realistic scenarios.

However, KLD as a distance metric fails to measure the quality of localization as directly as IoU (i.e., a value between 0 and 1). Thus, we perform a nonlinear transformation on KLD (i.e., KLD $= (1/1 + \text{KLD})$) to complete the normalization. Compared to IoU, KLD is insensitive to minor offsets of small bounding boxes and is able to measure non-overlapping objects. Following [41], to more accurately measure the localization quality of the tiny objects, we replace $\mathcal{N}_p$ with the receptive field of the anchor.

B. Adaptive Online Training Sample Mining

Label assignment with a splitting condition of IoU or centroid distance achieves favorable performance and is gradually becoming the de facto standard in object detection. Nevertheless, such a harsh split leads to imbalanced assignment results when dealing with tiny objects that are sensitive to localization metrics [i.e., the number of positive samples for normal-scale objects is much larger than that for tiny objects, as shown in Fig. 4(a)]. In this part, we propose an Ali-TSM strategy to break the limitation of pre-defined thresholds, allowing detectors to select high-quality samples in a flexible manner.

Considering the prediction results with simultaneously high classification scores and localization quality in the inference process are more likely to be retained, we define anchors with such property as high-quality samples. So, to better align with the inference, we pay more attention to these samples in the training process. To do this, we design a new quality score to clarify the importance of anchors

$$q = \text{cls_}c \times \text{loc_}a^{\beta} \tag{10}$$

where cls_c and loc_a (measured by KLD) denote the classification confidence and localization accuracy, respectively, and

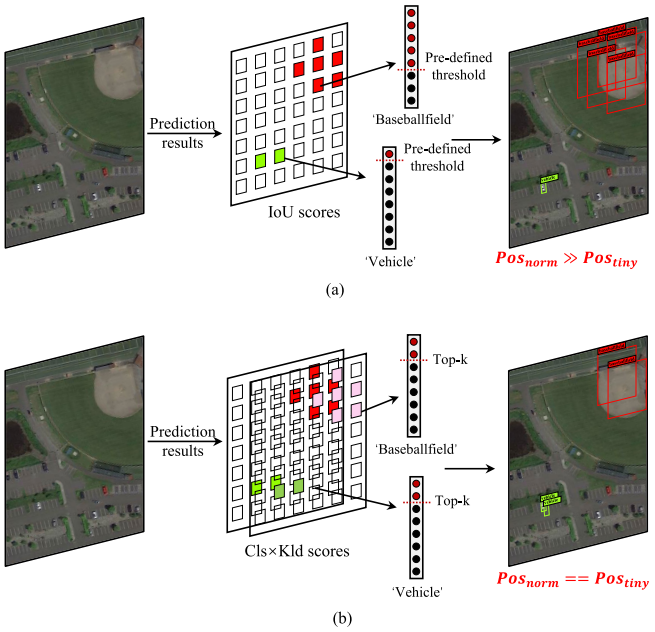


Fig. 4. Comparison of (a) label assignment based on pre-defined thresholds and (b) our method (Ali-TSM). Pos_{norm} and Pos_{tiny} denote the number of positive samples for normal- and tiny-scale objects, respectively. Our strategy assigns the same number of high-quality positive samples to objects across different scales.

β is a control parameter, balancing the effects of localization and classification. Notably, q plays a crucial role in aligning training and inference, as it encourages the network to dynamically select high-quality samples from a joint optimization perspective. Generally speaking, the GT box of a tiny object hardly overlaps with most prior boxes or points, causing the assigner to fail to assign labels to candidate boxes based on the ranking [41]. Fortunately, KLD is able to consistently respond to the matching degree of candidate boxes to GT, even if they do not overlap.

As discussed in [38], training sample selection is critical to the training of the detector. To avoid imbalanced assignment results, we propose a simple assignment rule: for each GT, the top k anchors with maximum q are selected as positive samples. As shown in Fig. 4(b), our method makes the model pay the same attention to instances across different scales.

To further expand the number of positive samples and to fit the normal-scale objects that appear in realistic scenarios, we mine additional samples by the following strategy: 1) retaining IoU as a candidate for the localization metric as it is friendly to normal-scale objects and 2) introducing a new quality score to ensure the diversity of positive samples. Based on this, we introduce cls_c $\times$ $(1-\alpha)+$iou $\times$ α as the quality score during sample mining. Here, we add only one additional positive sample to avoid introducing noise. As an empirical rule [37], α is set to 0.75.

C. Task-Level Attention

As mentioned above, it is still an open problem to modify the *Head* to precisely locate tiny objects with extremely limited pixels. As a result, we perform TlA on the output features of FPN to satisfy the demand for precise localization of tiny objects. TlA is conceptually simple: 1) we first embed

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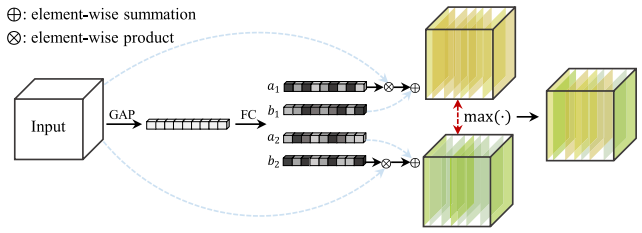


Fig. 5. Illustration of TIA. The input is derived from the output features of FPN.

global information by simply using global averaging pools to generate channel statistics (e.g., $\mathcal{F}_{\text{avg}} \in \mathbb{R}^C$); 2) we then generate weight vectors via two fully connected layers; and 3) we dynamically switch ON and OFF channels of features to support different tasks.

Given a feature $\mathcal{F} \in \mathbb{R}^{H\times W\times C}$ of a certain layer of FPN, the global context $\mathcal{F}_{\text{avg}} \in \mathbb{R}^C$ is obtained by global averaging pooling

$$\mathcal{F}_{\text{avg}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathcal{F}_{i,j} \tag{11}$$

where $\mathcal{F}_{\text{avg}}$ embeds the channel importance of the input features. Then, $\mathcal{F}_{\text{avg}}$ is fed into an activation layer consisting of two fully connected (fc) layers and activation functions to obtain the channel weights

$$[a_1, b_1, a_2, b_2]^T = \sigma\left(\text{fc}_2\left(\gamma\left(\text{fc}_1\left(\mathcal{F}_{\text{avg}}\right)\right)\right)\right) \tag{12}$$

where σ and γ denote HSigmoid$(\cdot)$ and ReLU$(\cdot)$, respectively. $\text{fc}_1(\cdot)$ is used to compress the dimensions, while $\text{fc}_2(\cdot)$ is used to expand the dimensions to $4C$. The nonlinear operation introduced by (12) makes the weight vector more representative.

As shown in Fig. 5, TIA supports the two tasks by toggling the switches of feature channels

$$\mathcal{F}^{\text{cls/reg}} = \max(a_1 \cdot \mathcal{F} + b_1,\ a_2 \cdot \mathcal{F} + b_2) \tag{13}$$

where $\max(\cdot)$ is acting on the feature slices of each channel. Equation (13) encodes the global context into the weight vector and adjusts the activation function accordingly to obtain stronger representations. Our TIA is able to freely explore different representations of objects via switchable functions, thus providing powerful support for diverse tasks (i.e., classification and regression). TIA is implemented similar to [14], but to the best of our knowledge, we first applied this concept in RSIs to facilitate the localization of tiny objects.

D. Loss Function

Our KLDet is implemented on the one-stage anchor-free detector FCOS [31]. In this article, the total loss consists of a multitask loss function for the classification and regression branches

$$L_{\text{total}} = \frac{1}{N_{\text{pos}}} \sum_{x,y} L_{cls}\left(\boldsymbol{p}_{x,y}, c^*_{x,y}\right) + \frac{\lambda}{N_{\text{pos}}} \sum_{x,y} \mathbb{1}_{\{c^*_{x,y}>0\}} L_{\text{reg}}\left(\boldsymbol{t}_{x,y}, \boldsymbol{t}^*_{x,y}\right) \tag{14}$$

where $\boldsymbol{p}_{x,y}$ and $\boldsymbol{t}_{x,y}$ denote the predicted classification scores and bounding box coordinates, respectively. N_{pos} denotes the number of positive samples and λ is the control factor (set to 1) for balancing L_{reg}. $\mathbb{1}_{\{c^*_{x,y}>0\}}$ is the indicator function, being 1 if the GT's category $c^*_{x,y} > 0$ and 0 otherwise (background class). $\boldsymbol{t}^*_{x,y} = [l^*, t^*, r^*, b^*]$ is the regression target at position (x, y). Formally, if location (x, y) is associated with the ith GT, the regression target for location can be formulated as

$$\begin{aligned} l^* &= x - x_0^{(i)}, \quad t^* = y - y_0^{(i)} \\ r^* &= x_1^{(i)} - x, \quad b^* = y_1^{(i)} - y \end{aligned} \tag{15}$$

here, $(x_0^{(i)}, y_0^{(i)})$ and $(x_1^{(i)}, y_1^{(i)})$ denote the coordinates of the upper-left and right-bottom corners of the GT bounding box. L_{cls} is original focal loss (FL) [13], which is used to solve the problem of extreme imbalance between foreground and background during the training process, and its typical form is as follows:

$$L_{cls} = \begin{cases} \left(1 - \boldsymbol{p}_{x,y}\right)^{\gamma} \log\left(\boldsymbol{p}_{x,y}\right), & \text{if } c^*_{x,y} = 1 \\ \left(\boldsymbol{p}_{x,y}\right)^{\gamma} \log\left(1 - \boldsymbol{p}_{x,y}\right), & \text{otherwise} \end{cases} \tag{16}$$

where γ is a tunable focusing parameter ($\gamma = 2$) that is used to reduce the contribution of simple examples and automatically focuses the model on hard examples. L_{cls} can better handle the imbalance between positive and negative samples, and pay more attention to difficult-to-distinguish tiny objects, thereby making the model's category predictions for tiny objects more reliable. L_{reg} is Distance-IoU Loss [49], and its formula is defined as follows:

$$L_{\text{reg}} = 1 - \text{IoU} + \frac{\rho^2\left(\boldsymbol{b}, \boldsymbol{b}^{gt}\right)}{c^2} \tag{17}$$

where $\boldsymbol{b}$ and $\boldsymbol{b}^{gt}$ denote the current position and its corresponding GT. $\rho(\cdot)$ is the Euclidean distance, and c is the diagonal length of the smallest external box that covers the two boxes. The distance penalty term $(\rho^2(\boldsymbol{b}, \boldsymbol{b}^{gt})/c^2)$ of L_{reg} provides better constraints on the positional relationship of the bounding box, thus improving the localization accuracy of tiny objects.

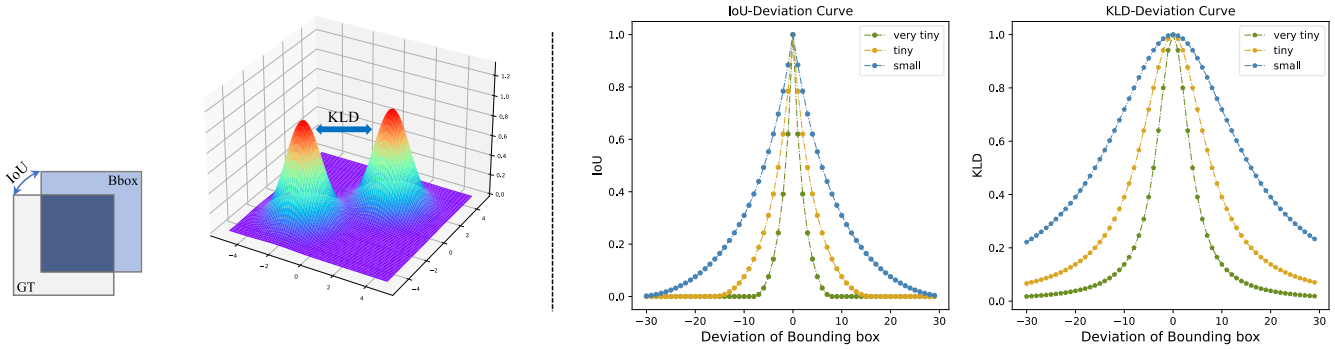
IV. Experiment

In this section, we first describe four public remote sensing object detection datasets, evaluation metrics, and implementation details. Then, all experiments are analyzed quantitatively and qualitatively to validate the effectiveness of our KLDet.

A. Datasets

1) Tiny Object Detection in Aerial Images [2]: AI-TOD is an optical remote sensing dataset designed for the detection of tiny objects. It contains eight common categories, 28 036 RSIs of 800 × 800 pixels with 700 621 object instances. The instances in an image are classified into four sub-scales according to their absolute pixel areas: very tiny (*vt*) with $2^2 <$ area $\leq 8^2$, tiny (*t*) with $8^2 <$ area $\leq 16^2$, small (*s*) with $16^2 <$ area $\leq 32^2$, medium (*m*) with $32^2 <$ area $\leq 64^2$, and no large objects. Different scale subsets are illustrated in Table I. The average absolute size of AI-TOD is only 12.8 pixels, which is much smaller than the natural scene datasets MS COCO (99.5 pixels) [50] and PASCAL VOC (156.6 pixels) [51], and the remote sensing dataset DOTA (55.3 pixels) [52]. Some typical visualization results on AI-TOD are shown in Fig. 7.

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(a) Compare the calculation of different localization metrics

(b) Comparing the deviation curves of different localization metrics

Fig. 6. Comparison of different bounding box representations and deviation curves of their localization metrics. In (a), the curve with double arrows on the left indicates the IoU between the bounding boxes, and the arrow on the right refers to the distance between the corresponding two distributions. In (b), the horizontal axis value indicates the number of pixel deviations at the center of bounding boxes, and the vertical axis value indicates the corresponding metric values. Best viewed in color.

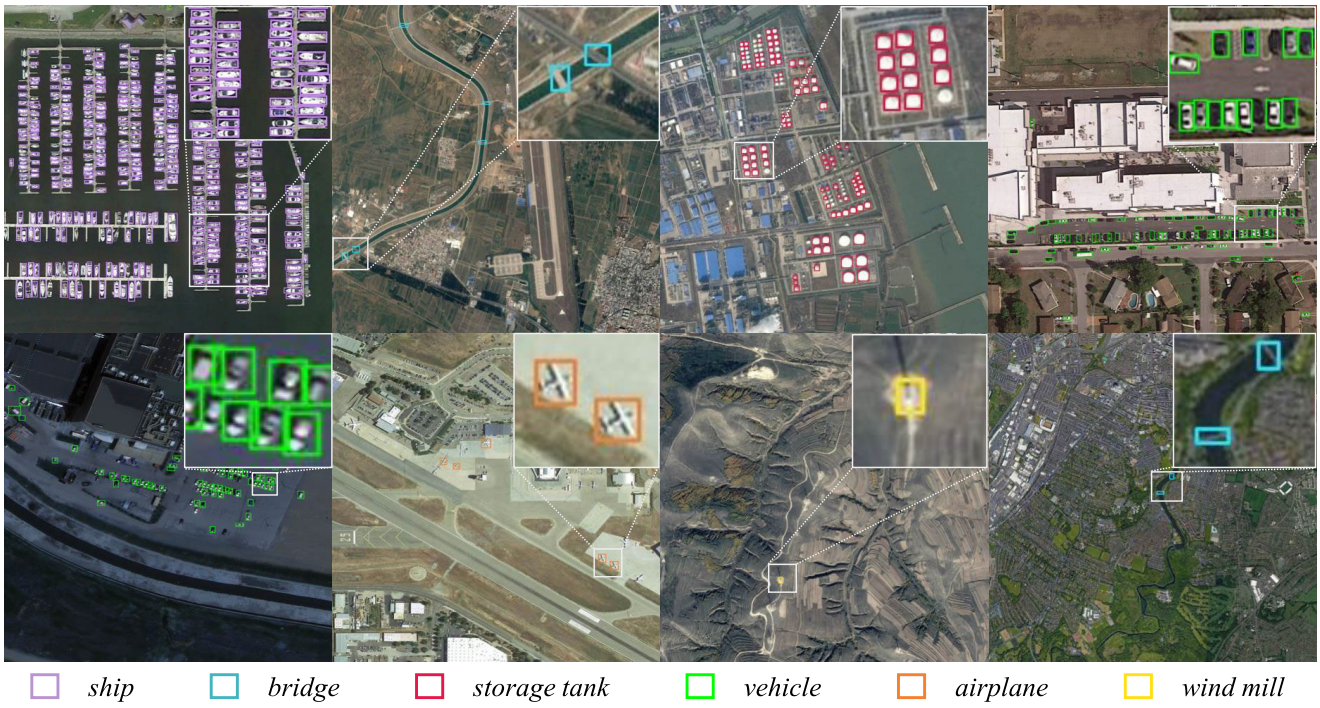


Fig. 7. Some representative detection results of KLDet on the AI-TOD dataset. Best viewed in color, and zoomed in for details.

TABLE I
SCALE SUBSETS AND CORRESPONDING AREA RANGES OF OBJECTS

Area Subset	*Tiny*		*Small*	*Medium*
	very tiny	*tiny*		
Area Range	$(2^2, 8^2]$	$(8^2, 16^2]$	$(16^2, 32^2]$	$(32^2, 64^2]$

2) DIOR [15]: DIOR is an ultra-large scale RSIs benchmark with 192 472 axis-aligned bounding box annotations across 23 463 RSIs of 800 × 800 pixels covering 20 object categories. To simplify, the 20 categories (c1-c20) are abbreviated to be written as follows: airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (BG), chimney (CM), dam (DM), expresswayservice-area (EA), expressway-toll-station (ES), golf course (GC), ground track field (GF), harbor (HB), overpass (OP), ship (SP), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (VH), and windmill (WM). Visualization results in DIOR are shown in Fig. 8.

3) VisDrone [16]: VisDrone contains numerous tiny objects under different imaging conditions due to variations in viewpoints and heavy occlusions. There are 10 209 images with 2000 × 1500 pixels and 542K instances covering ten common object categories in traffic scenes.

4) SODA-D [17]: It highlights the detection of those objects with extremely tiny size under complicated driving scenarios. Covering a wide range of weather, view and lighting conditions, SODA-D boasts a rich diversity of locations, weather, periods, shooting views and scenes. As a super-large dataset, SODA-D has 328 728 images of size 800 × 800 pixels.

B. Evaluation Metrics

Following the evaluation protocols in MS COCO [50], we use the average precision (AP) to evaluate the performance of detectors. The relevant formulas are listed below

$$\mathrm{AP} = \int_0^1 P(R)\, dR \tag{18}$$

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Fig. 8. Some representative detection results of KLDet on the DIOR dataset. Best viewed in color, and zoomed in for details.

here, P and R refer precision and recall. The precision P and recall R are defined as follows:

$$P = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FP}}} \tag{19}$$

$$R = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FN}}} \tag{20}$$

here, N_{TP}, N_{FP}, and N_{FN} denote the number of true positive (TP), false positive, and false negative (FN) samples, respectively. As the paramount metric, the overall AP is obtained by averaging the AP over ten IoU thresholds between 0.5 and 0.95 (with an interval of 0.05) on tiny objects.

There are several AP sub-metrics employed to compare the performance. Among them, AP denotes the overall accuracy, and AP_{50} and AP_{75} denote the accuracy at IoU thresholds of 0.5 and 0.75. In addition, to emphasize the robustness of the model to objects at different scales, we demonstrate the AP for different area subsets, such as AP_{vt}, AP_{t}, AP_{s}, and AP_{m} in AI-TOD, which represent the detection accuracy of very tiny, tiny, small, and medium-sized objects, respectively.

To reflect the computational complexity, we also provide metrics such as parameters (#Param.), floating-point operations (FLOPs), and frames per second (FPS). Where #Param. and FLOPs represent the number of model parameters and computational cost, respectively. FPS stands for inference speed, a larger value means a shorter test time for a single image.

C. Implementation Details

All experiments are performed on an NVIDIA A100 GPU, with implementations based on the MMDetection toolbox [53] and PyTorch [54]. As with most single-stage detectors [31], [32], we use a "backbone-FPN-head" detection pipeline with a ResNet-50 [44] backbone pre-trained on the ImageNet [55]. ResNet-50 as a light convolutional neural network architecture achieves a trade-off between accuracy and efficiency. For AI-TOD, all models are trained with a stochastic gradient descent (SGD) optimizer for 12 epochs ($1\times$ learning schedule) with 0.9 momenta, a batch size of 2, and an initial learning rate of 0.0025 and made to decay by a factor of 10 at the 8th and 11th. For DIOR, following the previous work [22], we train the model by 12 epochs, with an initial learning rate of 0.1 and reduced by a factor of 0.01 at the 8th and 11th epochs, and a batch size of 8. The focusing parameter γ in loss function is 2, as used in [13]. For better handling of tiny objects, we use P2-P6 of FPN [8]. During inference, a confidence score of 0.05 is set to filter the background, the IoU threshold of non-maximum suppression (NMS) is set to 0.5, and the first 3000 bounding boxes are saved to fit the dense

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TABLE II
EFFECT OF DIFFERENT HYPERPARAMETERS IN KLDET ON PERFORMANCE. k CONTROLS THE NUMBER OF TRAINING SAMPLES. β BALANCES CLS_c AND LOC_a IN CALCULATING QUALITY SCORE. THE BEST RESULTS ARE MARKED IN BOLD

k	β	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m	AP
2		41.1	11.5	7.4	18.6	19.2	22.4	17.2
3	5	43.0	12.1	7.5	19.4	21.4	23.7	18.1
4		40.7	12.4	7.6	19.0	20.2	23.7	17.6
3	4	43.7	12.3	7.9	19.6	21.3	23.8	18.2
	6	44.0	12.7	**8.2**	20.0	21.0	24.3	18.5
	7	**45.1**	12.5	7.7	**20.2**	**21.9**	24.4	**18.7**
	8	44.3	**13.0**	7.9	19.8	21.4	**25.0**	18.6

prediction of tiny objects, as used in [41]. Unless specified otherwise, the above parameters are used for all experiments.

The control parameter β in our KLDet is a hyperparameter balancing the quality of classification and localization, and k controls the number of training samples. A smaller k can select the best few boxes but may result in insufficient training, while a larger k can increase the training samples but may introduce noise. We perform a number of experiments to determine the appropriate values of β and k. In our experiments, we first fix β to 5 and choose k from 2 to 4. As shown in Table II, the turning point of accuracy occurs at $k = 3$ when $\beta = 5$. Therefore, we empirically set k to 3. Then, we change the value of β from 4 to 8. We find that the best combination of parameters is $\beta = 7$ and $k = 3$.

D. Comparison With State-of-the-Art Methods

In this section, the comparative experiments of KLDet and other state-of-the-art detectors on AI-TOD [2], DIOR [15], VisDrone [16], and SODA-D [17] are analyzed.

1) Results on AI-TOD: In Table III, we compare KLDet with the state-of-the-art methods on the AI-TOD dataset [2], and the results involved in these methods are reported in [3] and [41] and related articles. These methods include representative single- and two-stage generic detectors and detectors dedicated to remote-sensing objects. For generic detectors, the single-stage detector yields competitive performance at a lower cost compared with the two-stage detector. For example, VFNet [39] with ResNet-50 [44] achieves 13.9 AP (identical to Cascade R-CNN [33] configured with ResNet-101) and obtains better performance (29.3 versus 24.5) on the crowded tiny object SPs. This suggests that single-stage detectors have the potential to achieve the trade-off between accuracy and speed. Although the latest methods, such as NWD [3] and RFLA [41] dedicated to RSIs bring considerable gains to the two-stage detector, they achieve an absolute lead only on the large object ALs. In contrast, our KLDet beats most single- and two-stage detectors configured with NWD and RFLA, and brings higher gains for very tiny objects (e.g., 53.4 AP in SPs and 10.0 AP in WMs). Our KLDet outperforms the RFLA by a clear margin of 3.3 (19.6 versus 16.3) points when using the same baseline (i.e., FCOS [31]), indicating the importance of selecting high-quality training samples for the model. With the 2× training schedule, the performance of KLDet is further pushed to 25.2 AP, surpassing Faster RCNN [18] configured with RFLA for the first time and leading the baseline by 9.7 AP. The detection results are presented in Fig. 7. KLDet is ideal for detecting tiny objects, especially for BGs and WMs that almost blend into the background.

We also evaluate the accuracy of detectors on objects of different sizes. As shown in Table IV, though our method does not achieve the highest accuracy on the medium scale object, KLDet brings the maximum gain (2.8 in AP_{vt}, 3.8 in AP_t, 3.7 in AP_s, 4.0 in AP_m) compared to the baseline for different scale instances. ATSS equipped with NWD achieves the best performance on AP_m, which is due to the fact that the normal scale objects are not suitable for modeling as Gaussian distributions. In the average recall, the AR_s and AR_m of KLDet perform slightly worse compared to other methods. These results suggest that KLDet still has room for improvement on medium size objects.

2) Results on DIOR: To evaluate the robustness of our KLDet, comparison experiments are performed on the DIOR dataset [15], which is a challenging benchmark for detecting small objects in optical RSIs. As illustrated in Table V, our KLDet is highly competitive compared with other methods, achieving the most advanced performance in several categories (bold in red). For example, KLDet achieves state-of-the-art detection accuracy for "SPs," "STs," and "VHs" with extremely limited pixels. This result confirms that KLDet has significant advantages in detecting small objects. For two-stage detectors [18], [21], [35], they are constrained by the scale of the initial anchors and the heuristic-based label assignment, resulting in models that are only competitive for large objects but perform poorly on more challenging small objects. We also compare KLDet with the recently popular Transformer-based detectors [28], [29], [30]. However, they are quite weak in the optical RSIs dataset. For example, despite Deformable-DETR [28] is optimized for small objects and trained longer, our proposed KLDet still outperforms it 36.0 and 21.2 mAP on typical tiny objects "SPs" and "VHs." Compared to the remote sensing object detector FSoD-Net [23], although it achieves optimal accuracy in several categories (e.g., "BF" and "BC"), its improvements are focused on large-sized objects and overall performance is inferior to our proposed approach. Besides, a training schedule of 20 epochs is used for a fair comparison with ABNet [1]. As shown in the results, our proposed KLDet clearly beats ABNet and achieves 73.5 mAP. The visualization results are shown in Fig. 8. From the results, KLDet performs brilliantly in a variety of complex scenarios, especially for classifying and localizing small objects. For example, KLDet accurately detects "WMs" that are almost integrated into the background.

3) Results on VisDrone: In Table VI, we compare KLDet with the state-of-the-art methods on the VisDrone dataset [16]. The results involved in these methods are presented in [22] and related publications. Our KLDet outperforms all the mentioned one-stage/two-stage general detectors (e.g., DetectoRS [57] and ATSS [38]), indicating that our improvement is effective. Compared to existing state-of-the-art detectors dedicated to RSIs, including ClusDet [65], DMNet [66], RRNet [67], HawkNet [22], and RFLA [41], our KLDet achieves clear superiority in small object detection. ClusDet [65] focuses on enlarging small objects to enhance their detection

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TABLE III

QUANTITATIVE COMPARISON OF THE STATE-OF-THE-ART METHODS WITH OUR KLDET IN THE AI-TOD TEST SET. METHODS SPECIFICALLY DESIGNED FOR RSIS ARE MARKED WITH † FOR A FAIR COMPARISON. 2× INDICATES THE MODEL IS TRAINED USING A 2× LEARNING SCHEDULE (24 EPOCHS). * INDICATES OUR RE-IMPLEMENTATION BECAUSE THE ORIGINAL ARTICLE DID NOT CONDUCT THIS EXPERIMENT. THE RESULTS WITH BOLD RED AND BLUE COLORS INDICATE THE BEST AND SECOND-BEST RESULTS OF EACH COLUMN, RESPECTIVELY

Method	Backbone	Airplane	Ship	Person	Bridge	Swimming pool	Wind mill	Storage tank	Vehicle	AP
two-stage detectors:										
TridentNet [56]	ResNet-101	16.3	16.0	2.8	0.1	6.1	0.0	16.6	11.4	8.7
Faster R-CNN [18]	ResNet-50	21.8	19.9	4.1	2.5	8.2	0.0	19.9	12.6	11.1
Cascade R-CNN [33]	ResNet-50	13.3	24.8	6.4	5.1	3.3	2.6	29.1	21.4	13.3
Cascade R-CNN [33]	ResNet-101	23.8	24.5	5.6	9.8	11.8	0.3	20.5	15.3	13.9
DetectoRS [57]	ResNet-50	**27.1**	25.2	5.9	9.5	12.0	0.1	23.3	14.8	14.8
†Faster R-CNN [18] w. NWD [3]	ResNet-50	25.6	30.5	8.0	11.4	9.8	3.7	30.6	22.8	17.8
†Cascade R-CNN [33] w. NWD [3]	ResNet-50	24.7	34.5	8.0	12.0	11.6	3.3	32.6	23.1	18.7
†Faster R-CNN [18] w. RFLA [41]	ResNet-50	25.0	38.3	10.7	16.3	12.1	6.4	34.8	25.2	21.1
†Faster R-CNN*$_{2\times}$ [18] w. RFLA [41]	ResNet-50	**28.4**	**50.6**	**11.8**	17.4	**16.0**	**7.6**	**36.6**	**25.8**	**24.3**
single-stage detectors:										
RetinaNet [13]	ResNet-50	5.3	22.3	3.0	8.4	0.0	0.0	11.1	10.3	7.6
†RetinaNet [13] w. RFLA [41]	ResNet-50	0.0	29.9	4.3	12.2	0.0	0.0	11.5	14.9	9.1
AutoAssign [58]	ResNet-50	0.6	28.0	5.6	12.7	0.2	0.9	24.9	17.5	11.3
DDOD [59]	ResNet-50	15.8	26.3	5.7	3.3	0.0	0.0	26.4	20.4	12.2
ATSS [38]	ResNet-50	21.1	25.8	5.4	4.0	0.6	0.0	26.6	18.7	12.8
DyHead [14]	ResNet-50	20.7	27.1	5.9	4.4	1.7	0.2	25.6	18.9	13.1
†ATSS [38] w. NWD [3]	ResNet-50	21.1	25.8	5.7	3.9	5.0	0.8	27.0	18.3	13.5
VFNet [39]	ResNet-50	20.0	29.3	5.9	2.9	4.9	0.1	27.7	20.7	13.9
†AutoAssign [58] w. RFLA [41]	ResNet-50	12.2	27.3	6.5	10.3	6.3	2.6	28.9	19.3	14.2
GFL [60]	ResNet-101	23.1	32.1	6.7	3.2	4.8	0.0	29.7	21.5	15.2
FCOS [31]	ResNet-50	6.2	37.9	7.7	16.6	1.4	0.1	30.2	24.1	15.5
†FCOS [31] w. RFLA [41]	ResNet-50	0.9	40.1	8.1	17.0	5.0	5.6	30.2	23.8	16.3
KLDet (ours)	ResNet-50	13.6	42.3	9.3	**18.7**	5.2	6.7	35.7	24.9	19.6
†FCOS*$_{2\times}$ [31] w. RFLA [41]	ResNet-50	17.0	49.3	9.5	**18.7**	8.8	5.1	33.2	25.7	20.9
KLDet$_{2\times}$ (ours)	ResNet-50	22.9	**53.4**	**11.5**	**21.3**	**17.6**	**10.0**	**38.2**	**27.0**	**25.2**

TABLE IV

AP AND AR METRICS FOR DIFFERENT METHODS ON THE AI-TOD DATASET

Method	AP_{vt}	AP_t	AP_s	AP_m	AR_{vt}	AR_t	AR_s	AR_m	AP
ATSS [38]	1.9	11.6	19.5	29.2	2.9	25.7	41.2	44.9	12.8
ATSS w. NWD [3]	2.1	11.1	20.9	**31.9**	6.1	26.9	**39.4**	**45.0**	13.5
FCOS [31]	5.6	16.8	19.0	22.4	11.1	33.4	36.1	37.5	15.5
FCOS w. RFLA [41]	7.3	18.5	19.8	21.8	-	-	-	-	16.3
KLDet (ours)	**8.4**	**20.6**	**22.7**	26.4	**14.1**	**38.4**	37.8	42.5	**19.6**

performance, while RRNet [67] employs a re-sampling method to increase the number of objects in the image and improve overall object detection performance. On the contrary, KLDet does not require these complex operations to achieve competitive detection accuracy on small objects. To compare with advanced methods [64], [68], we train KLDet for 120 epochs by a multiscale strategy. HRGiraffe-Det [68] uses up-sampled images as input to retain more spatial information about small objects. SDP [64] emphasizes small object features by eliminating large object features in shallow layers. Benefiting from the well-designed label assignment strategy, our KLDet achieves the best results with an overall accuracy of 32.3 AP, and a small object detection accuracy of 23.9 AP_S under the multiscale training conditions. Additionally, the lower parameter of KLDet implies a smaller computational cost, making it easier to deploy in real-world scenarios.

4) Results on SODA-D: Results on the SODA-D dataset [17] are shown in Table VII. The results of the models involved in the comparison come from [17]. From the table, it can be seen that the overall accuracy of the baseline model FCOS [31] is 23.9 AP and performs poorly on extremely small objects (6.9 AP_{eS}). This may be that the label assignment based on the center region of GT in the anchor-free detector is not suitable for tiny objects with limited pixels. It can be noticed that even with more training times, CenterNet [69] and CornerNet [35] only obtain overall accuracies of 21.5 and 24.6 AP, which are significantly lower than our method. Sparse RCNN [34] achieves an accuracy of 24.2 AP, which is comparable to FCOS. We also compare our KLDet with the recently popular Transformer-based detector. Deformable-DETR [28] combines a sparse spatial sampling of deformable convolution [6] and relation modeling capability of Transformer [27]. It reduces high computation in encoder and improves detection accuracy for small objects by focusing on salient key elements in the feature map. However, even with more training epochs, Deformable-DETR only delivers 19.2 AP, which is significantly lower than our KLDet. This performance difference suggests that the sparse query paradigm may not adequately cover tiny objects. Notably, KLDet achieves competitive detection accuracy for extremely small objects (i.e., 10.7 AP_{eS}) at a lower computation and training cost.

E. Ablation Study

To verify the effectiveness of each component, we perform a series of comparative experiments on the `test set` of AI-TOD [2] and DIOR [15]. Table VIII illustrates the performance and cost of our detectors over different conditions.

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TABLE V

QUANTITATIVE COMPARISON OF THE STATE-OF-THE-ART METHODS WITH OUR KLDET IN THE DIOR TEST SET. 20E INDICATES THE MODEL TRAINED WITH 20 EPOCHS

Method	Backbone	AL	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM	mAP
two-stage detectors:																						
Dynamic RCNN [61]	ResNet-50	56.9	65.1	62.4	85.5	33.5	76.0	45.5	54.4	47.6	72.5	69.7	36.8	53.5	70.4	45.4	57.4	78.3	47.7	37.0	73.7	58.5
TridentNet [56]	ResNet-50	56.3	73.6	63.5	84.9	29.9	75.7	53.7	55.7	48.0	75.3	71.0	52.0	50.7	64.8	62.6	52.6	78.6	54.6	34.2	72.6	60.5
CornerNet [35]	Hour-104	58.8	**84.2**	72.0	80.8	46.4	75.3	64.3	81.6	**76.3**	79.5	79.5	26.1	60.6	37.6	70.7	45.2	84.0	57.1	43.0	75.9	64.9
Faster RCNN [18]	ResNet-101	54.0	74.5	63.3	80.7	44.8	72.5	60.0	75.6	62.3	76.0	76.8	46.4	57.2	71.8	68.3	53.8	81.1	59.5	43.1	81.2	65.2
Mask R-CNN [62]	ResNet-101	53.9	76.6	63.2	80.9	40.2	72.5	60.4	76.3	62.5	76.0	75.9	46.5	57.4	71.8	68.3	53.7	81.0	62.3	43.0	81.0	65.2
PANet [63]	ResNet-101	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.1
†CBD-E-FPN [20]	ResNet-101	54.2	77.0	71.5	87.1	44.6	75.4	63.5	76.2	65.3	79.3	79.5	47.5	59.3	69.1	69.7	64.3	84.5	59.4	44.7	83.1	67.8
†CSFF [21]	ResNet-101	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	**81.6**	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	68.0	68.0
single-stage detectors:																						
DAB-DETR [30]	ResNet-50	38.0	57.7	62.5	72.0	22.1	72.2	43.1	53.5	44.4	62.3	48.9	42.3	43.9	12.0	44.0	17.0	58.9	47.0	13.4	45.1	45.0
Deformable-DETR$_{50e}$ [28]	ResNet-50	70.9	55.8	65.0	82.7	23.4	75.9	37.1	37.8	45.4	55.1	56.2	9.7	44.9	55.8	38.0	44.4	79.7	36.6	36.1	69.5	51.0
DINO [29]	ResNet-50	47.8	71.5	62.9	78.7	28.7	72.7	53.3	55.9	45.3	67.3	55.9	53.9	49.3	16.9	48.6	32.0	71.0	52.4	17.1	52.8	51.7
TOOD [32]	ResNet-50	67.6	67.9	67.1	85.8	33.1	77.9	49.0	55.7	48.7	73.2	67.6	47.9	52.8	73.7	40.4	65.7	79.9	49.7	45.9	74.2	61.2
FCOS [31]	ResNet-50	54.8	77.2	73.6	87.3	36.8	**79.2**	56.2	75.7	60.7	**80.2**	75.6	50.2	55.2	69.7	62.9	51.1	85.7	51.2	41.2	83.5	65.4
RetinaNet [13]	ResNet-101	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
†O^2-DNet [25]	Hour-104	61.2	80.1	73.7	81.4	45.2	75.8	64.8	81.2	76.5	79.5	79.7	47.2	59.3	72.6	70.5	53.6	82.6	55.9	49.1	77.8	68.4
FoveaBox [36]	ResNet-50	66.9	79.6	76.7	87.6	42.7	**79.8**	60.6	81.1	66.4	74.5	80.3	50.0	57.6	73.1	71.5	60.0	86.9	54.5	42.7	88.3	69.0
†SCRDet++ [19]	ResNet-50	64.3	79.0	73.2	85.7	45.8	76.0	**68.4**	79.3	68.9	77.7	77.9	56.7	**62.2**	70.4	67.7	60.4	80.9	**63.7**	44.4	84.6	69.4
†FCOS* w. RFLA [41]	ResNet-50	69.3	75.3	76.9	87.2	43.5	75.0	62.2	77.8	68.3	77.6	79.3	51.6	57.3	91.1	62.5	74.9	87.4	54.8	**56.1**	89.9	70.9
†ASSD [24]	VGG16	**85.6**	82.4	75.8	**89.5**	40.7	77.6	64.7	67.1	61.7	**80.8**	78.6	**62.0**	58.0	84.9	**76.7**	65.3	87.9	**62.4**	44.5	76.3	71.1
†FSoD-Net [23]	ResNet-50	**88.9**	66.9	**86.8**	**90.2**	45.5	79.6	48.2	**86.9**	**75.5**	67.0	77.3	53.6	59.7	78.3	69.9	75.0	**91.4**	52.3	52.0	**90.6**	71.8
†HawkNet [22]	ResNet-50	65.7	**84.2**	76.1	87.4	45.3	79.0	64.5	82.8	72.4	50.2	**82.5**	**74.7**	59.6	89.7	66.0	70.8	87.2	61.4	52.8	88.2	72.0
KLDet (ours)	ResNet-50	67.8	78.5	**79.6**	87.0	46.2	78.6	61.3	79.9	71.8	76.6	80.3	54.3	59.8	**91.8**	60.5	**76.6**	**88.0**	56.2	**57.3**	89.2	72.1
†ABNet$_{20e}$ [1]	ResNet-50	66.8	**84.0**	74.9	87.7	**50.3**	78.2	**67.8**	**85.9**	74.2	79.7	81.2	55.4	**61.6**	75.1	74.0	66.7	87.0	62.2	53.6	89.1	**72.8**
KLDet$_{20e}$ (ours)	ResNet-50	67.5	81.1	75.5	88.2	**47.4**	78.6	65.0	82.9	72.7	79.9	80.8	57.1	61.2	**91.2**	**74.6**	**75.2**	87.4	56.9	56.0	**90.2**	**73.5**

TABLE VI

COMPARISON WITH STATE-OF-THE-ART MODELS ON THE VISDRONE DATASET. ALL THE MODELS ARE TRAINED WITH A RESNET-50 [44] AS THE BACKBONE EXCEPT FOR SDP (DARKNET53) [64]. "MS" DENOTES MULTISCALE TRAINING

Method	Epoch	AP	AP$_{50}$	AP$_{75}$	AP$_S$	AP$_M$	AP$_L$	#Param.
two-stage detectors:								
Cascade-RCNN [33]	36	23.2	39.9	23.4	16.5	36.8	39.4	68.95M
†ClusDet [65]	42	26.7	50.6	24.7	17.6	38.9	**51.4**	-
DetectoRS [57]	42	26.8	43.2	28.0	17.5	38.2	41.7	134.30M
Faster R-CNN [18]	120	27.9	47.0	29.0	17.9	40.6	44.3	41.16M
†DMNet [66]	42	28.2	47.6	28.9	19.9	39.6	**55.8**	-
†RRNet [67]	247	29.1	**55.8**	27.2	-	-	-	-
single-stage detectors:								
RetinaNet [13]	42	22.1	36.8	23.0	10.4	35.7	46.0	35.68M
FCOS [31]	42	23.7	39.9	24.6	14.6	34.4	42.0	31.92M
†FCOS w. RFLA [41]	42	24.3	43.5	23.6	16.1	33.5	37.8	31.92M
†HawkNet [22]	36	25.6	44.3	25.8	19.9	36.0	39.1	-
VFNet [39]	120	25.9	42.1	27.0	16.8	37.3	41.4	32.67M
ATSS [38]	120	27.6	45.5	28.5	18.1	39.2	42.1	31.32M
KLDet (ours)	42	28.5	50.7	28.1	20.0	37.9	41.7	32.00M
†HRGiraffe-Det$_{Ms}$ [68]	120	**30.2**	49.1	**31.5**	21.3	**42.0**	43.3	-
†SDP$_{Ms}$ [64]	120	**30.2**	52.5	30.6	**22.6**	39.6	39.8	-
KLDet$_{Ms}$ (ours)	120	**32.3**	**55.3**	**33.2**	**23.9**	**42.4**	42.5	32.00M

TABLE VII

COMPARISON WITH STATE-OF-THE-ART MODELS ON THE SODA-D TEST SET. ALL THE METHODS ARE SINGLE-STAGE DETECTORS AND TRAINED WITH RESNET-50 [44] AS THE BACKBONE, EXCEPT FOR CORNERNET (HOURGLASSNET-104) [35]

Method	Epoch	AP	AP$_{50}$	AP$_{75}$	AP$_{eS}$	AP$_{rS}$	AP$_{gS}$	AP$_N$	#Param.
Deformable-DETR [28]	50	19.2	44.8	13.7	6.3	15.4	24.9	34.2	35.17M
CenterNet [69]	70	21.5	48.8	15.6	5.1	16.2	29.6	**42.4**	70.75M
FCOS [31]	12	23.9	49.5	19.9	6.9	19,4	30.9	40.9	31.92M
Sparse RCNN [34]	12	24.2	**50.3**	20.3	**8.8**	20.4	30.2	39.4	105.96M
CornerNet [35]	24	**24.6**	49.5	**21.7**	6.5	**20.5**	**32.2**	**43.8**	200.96M
KLDet (ours)	12	**25.9**	**53.8**	**21.4**	**10.7**	**22.2**	**31.9**	41.6	32.00M

The baseline (FCOS [31]) uses ResNet-50 as the backbone, balances positive, and negative samples by focal loss [13], and centerness branch to evaluate the accuracy of prediction. In the following experiments, all hyperparameters are consistent for a fair comparison.

1) Effect of Bbox2Gaussian: To verify the effectiveness of Bbox2Gaussian, we compare the performance gain with and without it in Table VIII. The results indicate that it improves the detection accuracy of the model on AI-TOD and DIOR by 1.1 AP and 3.8 mAP, respectively, without introducing any additional parameters. Compared to the baseline (i.e., parameterized bounding box), Bbox2Gaussian models the boxes as Gaussian distributions and evaluates the quality of localization by measuring the similarity between the distributions. Thus, it is possible to replace the de facto standard localization metric IoU/Region with KLD. As shown in Fig. 6(b), although IoU is more friendly to normal-scale objects, its deviation-sensitive nature makes it easier to ignore the positive samples of the already few tiny objects. On the contrary, in dealing with tiny objects with extremely limited pixels, our localization metric provides the basis for selecting high-quality samples.

2) Effect of Ali-TSM: The Ali-TSM proposed in KLDet is designed to assign the same number of high-quality positive samples to GTs of different scales. As shown in Table VIII, Ali-TSM outperforms the baseline 3.3 AP (18.7 versus 15.4) and 6.2 mAP (71.6 versus 65.4) at cost-free, which demonstrates the effectiveness of this component. As previously analyzed, due to the inherent flaws of pre-defined thresholds, the label assignment strategy of the baseline causes an imbalance among positive samples.

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TABLE VIII

ABLATION STUDY ON THE AI-TOD DATASET [2] AND THE DIOR DATASET [15]. THE NUMBERS IN () INDICATE IMPROVEMENTS RELATIVE TO THE BASELINE

Bbox2Gaussian	Ali-TSM	TlA	AP on AI-TOD (%)	mAP on DIOR (%)	#Param. (M)	FLOPs (G)	Average computation time per image (s)
			15.4	65.4	31.92	339.24	0.052
✓			16.5(+1.1)	69.2(+3.8)	31.92	339.24	0.052
✓	✓		18.7(+3.3)	71.6(+6.2)	31.92	339.24	0.052
		✓	18.0(+2.6)	66.6(+0.8))	32.00	339.26	0.055
✓	✓	✓	**19.6(+4.2)**	**72.1(+6.7)**	32.00	339.26	0.055

TABLE IX

EFFECT OF TLA ON LOCALIZATION ACCURACY. THE BASELINE MODEL IS FCOS [31]

Method	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP
Baseline	36.3	10.9	7.3	18.5	15.4
TlA (ours)	**43.1 (+6.8)**	**12.1 (+1.2)**	**7.8**	**19.2**	**18.0**

TABLE X

IMPACT OF DIFFERENT BOUNDING BOX REPRESENTATIONS ON PERFORMANCE

Type	Metric	AP_{vt}	AP_t	AP_s	AP_m	AP
Parameterized	IoU	2.6	10.0	17.8	**25.5**	11.6
	Region	5.0	17.6	18.5	20.7	15.4
Distribution (ours)	KLD	**7.7**	**18.8**	**19.6**	21.6	**16.5**

TABLE XI

COMPARISON BETWEEN DIFFERENT SCHEMES OF TRAINING SAMPLE ASSIGNMENTS. "POS/NEG": ASSIGNMENT CONDITIONS FOR POSITIVE/NEGATIVE SAMPLES. "REGION": ANCHORS FALLING INTO GT ARE POSITIVE SAMPLES. α IS 0.75, AS IN [37]

Method	Pos/neg	AP_{vt}	AP_t	AP_s	AP_m	AP
Region	*fixed*	5.0	17.6	18.5	20.7	15.4
$\text{Cls} \cdot \text{IoU}^{\beta}$	*top-k*	7.4	18.9	20.1	**25.5**	17.6
$(1-\alpha) \cdot \text{Cls} + \alpha \cdot \text{KLD}$		7.2	19.2	21.5	24.7	17.7
$\text{Cls} \cdot \text{KLD}^{\beta}$(ours)		**7.7**	**20.2**	**21.9**	24.4	**18.7**

3) Effect of TlA: Given the need for precise localization of tiny objects, we introduce TlA to guide the parallel branches in *Head* to select appropriate features that better support classification and regression tasks. Our TlA gains the baseline by 2.6 AP on AI-TOD [2] (18.0 versus 15.4) by adding about 0.08M parameters and 0.02G FLOPs. This means that TlA can be adopted in single-stage detectors in a plug-and-play manner, and provides better support for different tasks with almost no parameters.

To further verify this, we report the overall accuracy with AP metrics at different regression accuracy thresholds (0.50, 0.75). The results are shown in Table IX. Typically, a higher AP indicates a higher localization accuracy of the model when the regression threshold is fixed. Under the loose standard, i.e., an IoU threshold of 0.5 for judging a TP, TlA outperforms the baseline detector with a clear margin of 6.8 AP_{50} (43.1 versus 36.3). It validates our hypothesis that flexible cross-channel interactions can provide powerful support for different tasks, yielding more accurate detection. Benefiting from the introduction of TlA, our method brings a 1.2 AP_{75} boost to the baseline even under the more rigorous condition with 0.75 as the threshold.

F. Model Analysis

1) Comparison of Different Bbox Representations: As illustrated in Fig. 6, we qualitatively compare the influence of parameterized boxes and Gaussian distribution-based boxes on label assignment. In Fig. 6(b), we keep GT and Bbox in the same scale and move Bbox along the diagonal of GT to get the metric value. As illustrated in the results, the IoU values are very sensitive to deviations of tiny objects, and matching a strict division conditional on a pre-defined IoU threshold leads to an imbalance of positive samples across scales. In contrast, for very tiny objects with extremely limited pixels, KLD can measure the localization quality of the Bbox to some extent even if the Bbox does not intersect the GT, thus providing more potential positive samples for the model. Additionally, a smooth deviation curve suggests that even under strict rigid division conditions, the KLD-based metric makes it easier to select positive samples. The quantitative comparison of different bounding box representations in AI-TOD is shown in Table X. For parameterized bounding boxes, typical anchor-based detectors usually select pos/neg samples based on IoU, while anchor-free detectors mainly classify pos/neg samples by determining whether the point prior falls in the central region of GT. From the results, it is clear that for tiny objects, IoU-based label assignment strategies are superior only for normal-scale objects, since they cannot correctly measure the minor offsets of tiny objects. The region-based strategy brings more positive samples for tiny objects, but has limited improvement in performance. It is worth noting that KLD brings 5.1 AP_{vt}, 8.8 AP_t, and 1.8 AP_s gains to the model compared to IoU metric, and these improvements are mainly focused on tiny objects, indicating that the distribution-based representation of the bounding box is more friendly to tiny objects.

2) Comparison of Different Label Assignments: To illustrate the effectiveness of Ali-TSM, we compare it with other label assignment schemes. As shown in Table XI, the fixed assignment strategy only reached 15.4 AP. In adaptive assignment strategies, we choose the cleanliness score [37] (the third row) as an alternative form of quality score for comparison with our method. Besides, we attempt different localization metrics (i.e., IoU) in the structure of quality scores to determine the most suitable combination. We first combine the classification score and IoU, although it is suboptimal for small objects the IoU property makes it work best for medium-scale objects. Cleanliness scores designed for generic object detection do not work well with tiny objects. In contrast, the quality score of our design brings the greatest improvement to the model

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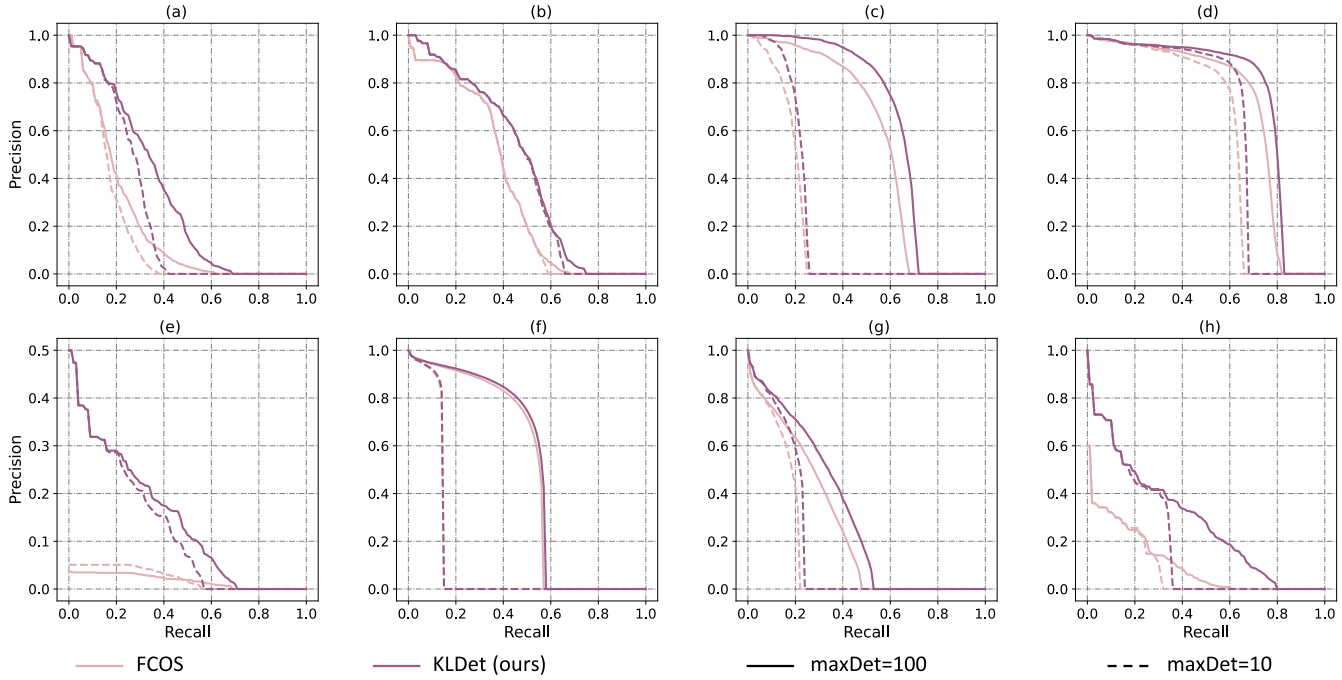


Fig. 9. P-R curves for our KLDet and baseline FCOS for different IoU thresholds, categories, and maximum number of predictions per image (maxDet). Best viewed in color. (a) PR curve @0.5, AL. (b) PR curve @0.5, BG. (c) PR curve @0.5, ST. (d) PR curve @0.5, SP. (e) PR curve @0.5, swimming-pool. (f) PR curve @0.5, VH. (g) PR curve @0.5, person. (h) PR curve @0.5, WM.

(18.7 versus 15.4 AP). As shown in Fig. 4, the key to the success of our Ali-TSM is to get rid of the restriction of pre-defined thresholds and turn to the ranking scheme to ensure that objects at different scales receive the same degree of supervision. This strategy alleviates the issue of insufficient training samples for tiny objects and enables the model to have more reasonable supervision for instances of various sizes.

3) Effect of Sample Mining: Sample mining can introduce a more diverse set of training samples, covering a wider range of scenes and variations in objects. In real-world scenarios, objects of different sizes are commonly encountered. Focusing solely on tiny objects may compromise the robustness of the model to some extent. To address the limitations of KLD in accurately localizing normal-sized objects, we introduce a sample mining strategy in Ali-TSM to supplement additional training samples. Specifically, we select additional training samples based on the score: $\text{cls_}c\times(1-\alpha)+\text{iou}\times\alpha$, where α is a hyperparameter that balances classification and regression. The motivation comes from: IoU can accurately reflect the localization precision of normal-sized objects. By increasing the training sample of normal-sized objects, Ali-TSM makes the supervision degree of the model for different sized objects more reasonable, as it no longer focuses only on tiny objects. We quantitatively compare model performance with and without sample mining. As shown in Table XII, after introducing the sample mining strategy, we bring a gain of 1.6 AP_m for normal-scale objects (i.e., medium-scale in AI-TOD). The experimental results demonstrate the effectiveness of sample mining. We also note that sample mining enhances the model's localization ability, i.e., the model performance boosts by 1.1 AP_{50} with an IoU threshold of 0.5.

TABLE XII
COMPARISON OF MODEL PERFORMANCE WITH AND WITHOUT SAMPLE MINING

Method	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
Ali-TSM wo. mining	19.1	45.3	13.6	8.3	20.1	**23.4**	24.8
Ali-TSM w. mining (ours)	**19.6**	**46.4**	**13.7**	**8.4**	**20.6**	22.7	**26.4**

G. Discussion

1) Precision Versus Recall Curves on AI-TOD: To better understand how our KLDet improves detection performance, we present the precision versus recall (P-R) curves for each category on the `test set` of AI-TOD according to the official tool. Observing the overall shape of P-R curves provides an intuitive impression of the model's performance. A better model will typically maintain higher precision at higher recall, i.e., a greater area under curve (AUC). As illustrated in Fig. 9, our KLDet consistently improves performance under different conditions, e.g., category, object sizes, and maximum number of predictions per image during evaluation. Notably, our KLDet achieves significant accuracy gains at all recall ratios for both dense (e.g., STs, SPs, VHs) and sparse (e.g., BGs, WMs) objects. As illustrated in Fig. 9(e) and (h), KLDet brings impressive improvements to the swimming-pool and WM, i.e., their P-R curves are significantly refined. Based on this analysis, we believe that our proposed KLDet has a huge potential for constructing a detector that is applicable in realistic scenarios.

2) Confusion Matrix on DIOR: For DIOR, where instance sizes vary more dramatically, we compare the confusion matrix of KLDet and FCOS to illustrate the improvement of our method for each category. As shown in Fig. 10, the rows of the confusion matrix indicate the GT labels

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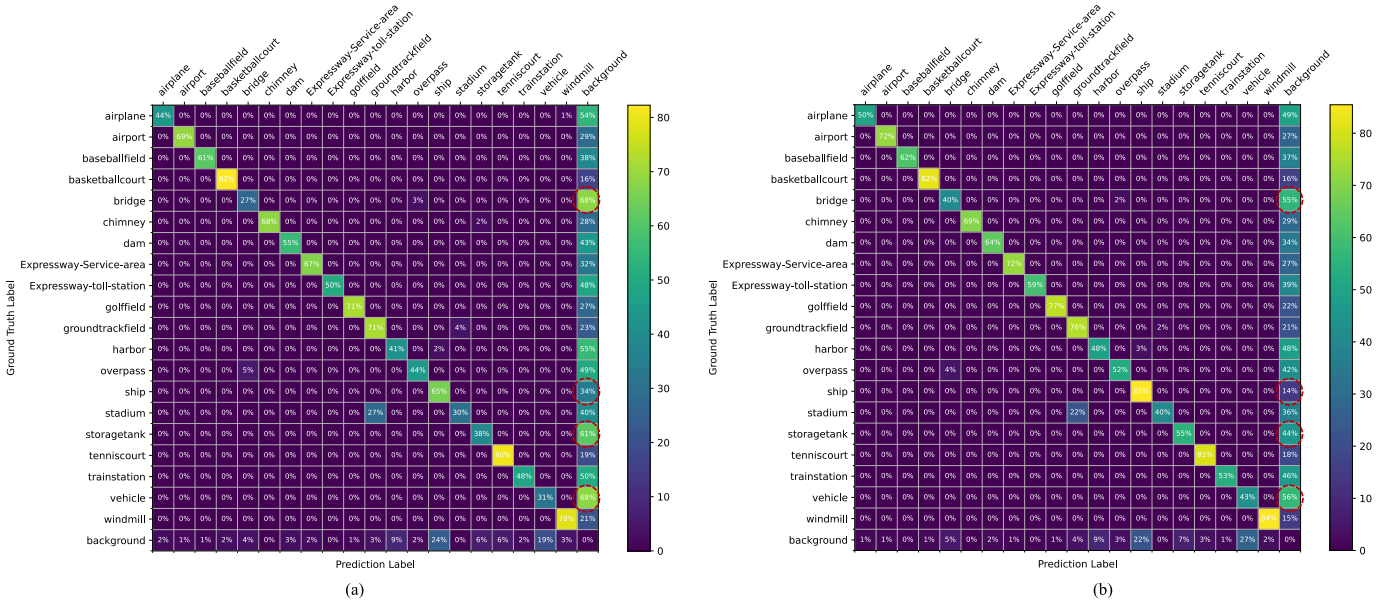
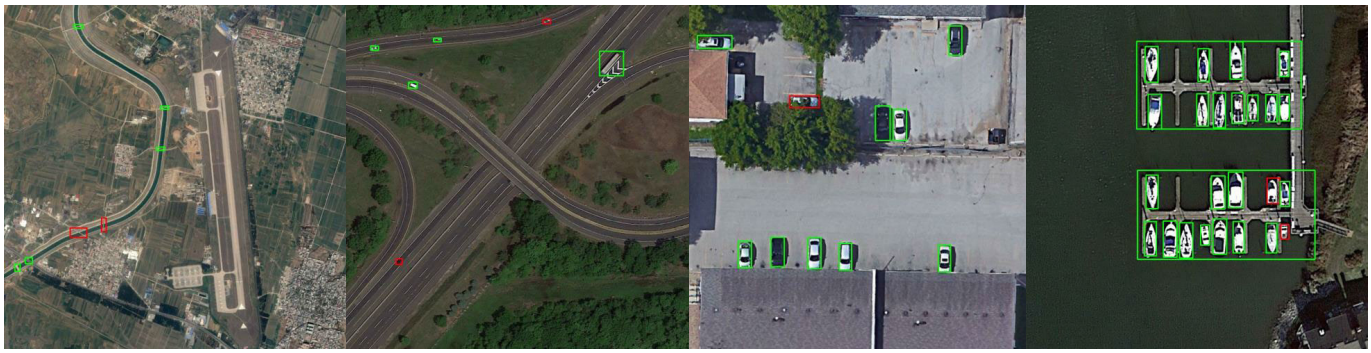


Fig. 10. Confusion matrix comparison of our KLDet with baseline FCOS on the DIOR dataset. Categories with sharp shifts in metrics are marked with red dashed lines. Best viewed in color. (a) FCOS. (b) KLDet (ours).



(a) (b) (c) (d)

Fig. 11. Some typical failure predictions of KLDet on AI-TOD and DIOR datasets. The two results on the left are from AI-TOD and the rest are from DIOR. The green and red bounding boxes indicate TP and FN predictions, respectively. (a) BG. (b) VH. (c) VH. (d) SP, HB.

and the columns indicate the predicted categories. Therefore, the last column indicates the missed detection rate for each category, while the bottom row indicates the false detection rate for different categories. The values on the diagonal of the confusion matrix represent the accuracy of predictions for each category. As illustrated in the results, there is a relatively high precision of FCOS on most normal-scale objects, but a large number of missed detections. For example, for BGs that almost blend into the background, densely aligned STs and VHs, the FCOS missed detection rate is between 60% and 70%. Lower accuracy and recall mean more errors and misses in the prediction results. In contrast, our KLDet reduces the rate of missed detections in these categories by approximately 15%, while significantly boosting their precision. This means that KLDet's predictions for various objects are more reliable and comprehensive. Unfortunately, although our method improves the precision, it slightly increases the false detection rate in certain categories, such as we detect some of the backgrounds as VHs.

3) Runtime Comparison: We also measure the training and testing times for different methods. For a fair comparison, the batch size is set to 4, the shorter side of the input images is fixed at 800 pixels, and all results are obtained on a single Tesla V100 GPU. As shown in Table XIII, where DetectoRS [57], FCOS [31], and Deformable-DETR [28] represent typical two-stage, one-stage, and transformer-based methods, respectively. The training time is measured on a single epoch. FCOS [31] is the baseline model and has a faster training and inference speed (∼0.5 H and ∼19.2 FPS). As a typical two-stage model, the refinement process of DetectoRS [57] makes it expensive for training and testing. Deformable-DETR [28] introduces sparse sampling to address the convergence issue of DETR [70]. Despite obtaining a faster training time (∼ 0.6 H), Deformable-DETR inference is only 15.3 FPS. Our KLDet maintains fast training and testing speeds of ∼0.6 H and ∼18.2 FPS, respectively, after introducing an acceptable computational cost.

TABLE XIII

COMPARISON OF TRAINING AND TESTING TIMES FOR DIFFERENT METHODS. H: HOURS, FPS: FRAMES PER SECOND

Time cost	DetectoRS [57]	FCOS [31]	Deformable-DETR [28]	KLDet (ours)
Train. (H)	1.1	**0.5**	0.6	0.6
Test. (FPS)	7.9	**19.2**	15.3	18.2

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4) Limitation of KLDet: Despite significant improvements brought by our KLDet as the baseline, object detection in optical RSIs remains an extremely challenging task. Inherent difficulties arise from various factors, such as the similarity in appearance between different objects and the intricate nature of the surrounding environment. In this part, in order to objectively learn the limitations of KLDet, we present some typical failures. As illustrated in Fig. 11(a) and (b), due to the blurred texture features, KLDet struggles to distinguish BGs and VHs that are closely connected to the road. Fig. 11(c) shows that KLDet may produce FN predictions when the object is obscured. Besides, Fig. 11(d) shows that some SPs may not be detected correctly. These cases imply that KLDet still has room for improvement in extracting discriminative features for tiny objects. Additionally, the lack of representative contextual features around the object can also contribute to such failed detections.

V. Conclusion

In this article, we propose a powerful single-stage detector, KLDet, which advances the detection of tiny objects in RSIs from the perspective of label assignment. First, we model the bounding box of tiny objects as a 2-D Gaussian distribution and utilize KLD as the localization metric, aiming to avoid suboptimal label assignment results caused by the IoU metric. Second, we develop an Ali-TSM strategy, which selects high-quality positive samples from both classification and localization. It also rationalizes the degree of supervision of the model over training samples of different scales. Finally, TlA is deployed to better support tiny object detection that requires precise localization. Notably, our method outperforms the baseline, especially for tiny object detection, by introducing only 0.08M additional parameters. On the AI-TOD, VisDrone, and SODA-D datasets, KLDet outperforms the latest single-stage detectors. Our best model achieves a state-of-the-art performance of 73.5 mAP on the DIOR dataset. However, numerous experiments show that as detection accuracy increases, missed detections inevitably occur. In future work, we will continue to explore localization metrics suitable for tiny-scale objects and strive to minimize FN predictions.

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